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United States Patent	12409536
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Olvera; Eduardo et al.

Tool for powered sprayer

Abstract

The present disclosure is directed to a tool for operating a nozzle holder. The tool may include an elongated body extending between a first end and a second end opposite the first end, the second end defining a recess configured to selectively receive and hold a portion of the spray nozzle. The tool may include a nozzle storage receptacle formed in the elongated body between the first end and the second end, the nozzle storage receptacle configured to selectively receive and hold the spray nozzle. The nozzle storage receptacle defines an aperture that extends through the elongated body between a first lateral side of the elongated body and a second lateral side of the elongated body opposite the first lateral side.

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Appl. No.: 18/671063

Filed: May 22, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240308041 A1	Sep. 19, 2024

Related U.S. Application Data

continuation parent-doc US 17552529 20211216 US 12017331 child-doc US 18671063
us-provisional-application US 63134386 20210106

Publication Classification

Int. Cl.: B25B27/14 (20060101); B05B15/60 (20180101)

U.S. Cl.:

CPC B25B27/14 (20130101); B05B15/60 (20180201);

Field of Classification Search

CPC: B25F (1/02)

USPC: 81/177.4; 81/490

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/552,529, filed Dec. 16, 2021, which claims priority to U.S. Provisional Patent Application No. 63/134,386, filed Jan. 6, 2021, the entire contents of all of which are incorporated herein by reference.

FIELD OF THE DISCLOSURE

(1) The present invention relates to powered sprayers, and more particularly to tools for operating a nozzle of a powered sprayer.

BACKGROUND OF THE DISCLOSURE

(2) A powered sprayer can include a removable nozzle. A tool can facilitate removing or installing the nozzle in the powered sprayer.

SUMMARY OF THE DISCLOSURE

(3) The present disclosure provides, in one aspect, a tool for operating a nozzle holder, the nozzle holder configured to selectively and removably couple a spray nozzle to a spray device, the tool including: an elongated body extending between a first end and a second end opposite the first end, the second end defining a recess configured to selectively receive and hold a portion of the spray nozzle; and a nozzle storage receptacle formed in the elongated body between the first end and the second end, the nozzle storage receptacle configured to selectively receive and hold the spray nozzle; wherein the nozzle storage receptacle defines an aperture that extends through the

elongated body between a first lateral side of the elongated body and a second lateral side of the elongated body opposite the first lateral side.

(4) The present disclosure provides, in another aspect, a tool for operating a nozzle holder, the nozzle holder configured to selectively and removably couple a spray nozzle to a spray device, the tool including: an elongated body extending between a first end and a second end opposite the first end, the second end defining a recess configured to selectively receive and hold a portion of the spray nozzle; and a nozzle storage receptacle formed in the elongated body between the first end and the second end, the nozzle storage receptacle configured to selectively hold and receive the spray nozzle inserted from a first lateral side of the elongated body and an opposite, second lateral side of the elongated body.

(5) The present disclosure provides, in another aspect, a tool for operating a nozzle holder, the nozzle holder configured to selectively and removably couple a spray nozzle to a spray device, the tool including: an elongated body extending between a first end and a second end; an annular rim defined by the first end, the annular rim being configured to selectively engage the nozzle holder to release the spray nozzle; and a nozzle storage receptacle formed in the elongated body between the first end and the second end, the nozzle storage receptacle configured to selectively hold and receive the spray nozzle inserted from a first lateral side of the elongated body and an opposite, second lateral side of the elongated body.

(6) Other aspects of the disclosure will become apparent by consideration of the detailed description and accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view of a fogger according to an embodiment of the disclosure.

(2) FIG. 2 is a perspective cross-sectional view of a sprayer unit of the fogger of FIG. 1, taken along line 2-2 of FIG. 1.

(3) FIG. 3 is a perspective cross-sectional view of portions of the sprayer unit of FIG. 2, taken along line 3-3 of FIG. 1.

(4) FIG. 4 is a cross-sectional view of portions of the sprayer unit of FIG. 2, taken along line 2-2 of FIG. 1.

(5) FIGS. 5-7 are perspective views of a tool for operating a nozzle holder of the fogger of FIG. 1.

(6) FIG. 8 is a side view of the tool of FIG. 5.

(7) FIG. 9 is a cross-sectional view of the tool of FIG. 5, taken along line 9-9 of FIG. 5.

(8) FIGS. 10A-10C are cross-sectional views of portions of the sprayer unit of FIG. 2 and of the tool of FIG. 5, illustrating a process for removing a spray nozzle from the nozzle holder using the tool.

(9) FIGS. 11A-11C are cross-sectional views of portions of the sprayer unit of FIG. 2 and of the tool of FIG. 5, illustrating a process for installing the spray nozzle into the nozzle holder using the tool.

(10) FIG. 12 illustrates a pair of backpack straps that can be attached to a backpack for carrying a remote reservoir of the fogger of FIG. 1.

(11) Before any embodiments of the application are explained in detail, it is to be understood that the application is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the following drawings. The application is capable of other embodiments and of being practiced or of being carried out in various ways.

DETAILED DESCRIPTION

(12) FIGS. 1 and 2 illustrate a spray device configured as an electrostatic powered sprayer or

fogger **10** for spraying a fine mist of atomized liquid disinfecting solution to disinfect surfaces. The fogger **10** includes a handheld sprayer unit **12** fluidly coupled to a remote reservoir **14** via a hose **16**. The sprayer unit **12** includes a first motor **18** that drives a liquid pump **20** to draw the liquid solution from the reservoir **14** and direct the solution through an atomizing nozzle assembly **22**. The sprayer unit **12** also includes a second motor **24** that drives a blower **26** to direct an airflow past the nozzle assembly **22** to increase the throw of the atomized solution.

(13) The sprayer unit **12** further includes a housing **28** that includes a main housing **30** in which the motor **18** and pump **20** are supported, and a tube **32** that defines an airflow channel **34**. The main housing **30** defines a handle portion **35** that may be grasped by the user during use to manipulate the orientation and position of the sprayer unit **12**. The main housing **30** also defines a battery receptacle **37** that selectively couples to a removable and rechargeable battery pack (not shown) to provide DC electrical power to the sprayer **10**. In alternative embodiments, the sprayer **10** may be AC powered (e.g., plugged into a standard home electrical socket).

(14) With reference to FIGS. 2 and 3, the sprayer **10** also includes an electrostatic charging circuit **39** operable to impart an electrostatic charge in the atomized liquid droplets exiting the nozzle assembly **22** during operation of the sprayer **10**. The electrostatic charging circuit **39** includes a high voltage supply module **41** supported within the main housing **30**, a first or high voltage electrode assembly **43** positioned in-line between the pump **20** and the nozzle assembly **22**, and a second electrode assembly or grounding ring assembly **45** suspended in front of the nozzle assembly **22**. The high voltage supply module **41** receives power from the battery pack and supplies power to the high voltage electrode assembly **43**. The grounding ring assembly **45** is electrically coupled to the high voltage supply module **41** to complete the charging circuit **39**. During operation of the sprayer **10**, the liquid solution passes through the high voltage electrode assembly **43** and is directly charged by direct contact therewith. An electric field develops between the grounding ring assembly **45** and any conductive material (such as, e.g., the liquid solution) in electrical contact with the high voltage electrode assembly **43**. As the atomized spray droplets of the liquid solution exit the nozzle assembly **22**, the droplets pass through the interior of the grounding ring assembly **45** and through the electric field, thereby acquiring an electrostatic charge. The acquired charge enables the spray droplets to better adhere or “stick” to targeted surfaces.

(15) The nozzle assembly **22** is suspended within the tube **32** proximate a forward end **36** that defines an outlet opening **38**. The nozzle assembly **22** includes a quick-connect nozzle holder **40** and a removable, replaceable nozzle **42**. The nozzle **42** can be removed from the nozzle holder **40** (e.g., for cleaning) and can also be replaced with a different nozzle (e.g., having different spray characteristics). Specifically, the nozzle holder **40** includes a retractable outer ring **44** that is movable between a holding position (FIG. 4), in which the nozzle **42** is securely held within the nozzle holder **40**, and a release position (FIGS. 10B and 10C), in which the nozzle **42** is released and ejected from the nozzle holder **40**. In other embodiments (not shown), the nozzle holder **40** can receive and hold the nozzle by other means. For example, the nozzle holder **40** can include internal threads for threadably receiving an externally threaded nozzle. In another example, the nozzle holder **40** can receive the nozzle in a bayonet lock arrangement.

(16) Since the nozzle assembly **22** resides within the tube **32**, it must be accessed through the outlet opening **38** to remove or replace the nozzle **42**. Accordingly, FIGS. 5-9 illustrate a tool **46** for removing or installing the nozzle **42** at the nozzle holder **40**. The tool **46** has a generally elongated rod-like body **47** that includes a first end or removal end **48**, and an opposite second end or installation end **50**. The removal end **48** defines a pocket **52** (FIG. 9) and includes an annular rim **54** communicating with the pocket **52** and having a diameter corresponding to a diameter of the retractable outer ring **44** of the nozzle holder **40**. The installation end **50** defines a recess **56** having a size and shape corresponding to the size and shape of a front portion of the nozzle **42**. In the illustrated embodiment, the recess **56** has a hexagonal shape corresponding to the hexagonal shape

of the front portion of the nozzle 42. The recess 56 receives the front portion of the nozzle 42 (e.g., by nominal slip fit) to temporarily hold the nozzle 42 so that it may be subsequently installed into the nozzle holder 40 using the tool 46.

(17) The tool 46 also includes nozzle storage receptacles 58 formed in a central region of the body 47 between the removal and installation ends 48, 50. The storage receptacles 58 receive additional nozzles 42 for storage and safekeeping. In the illustrated embodiment, the storage receptacles 58 define respective apertures that extend all the way through the body 47 between opposite first and second lateral sides 57, 59 of the body 47. As such, nozzle 42 may be inserted into each storage receptacle 58 from the first lateral side 57 or from the second lateral side 59. In other embodiments (not shown), the storage receptacles may instead define recesses that are open to only one of the lateral sides 57, 59 and are closed to the other lateral side 57, 59.

(18) The tool 46 further includes a flexible arm or clip 60 extending from the body 47 that allows the tool 46 to be secured to an article such as a belt, a strap, a pocket, or the like. For example, FIG. 12 illustrates a pair of backpack straps 62 that can be attached to a backpack for carrying the remote reservoir 14 (FIG. 1). The backpack straps 62 include a pocket 64, and the tool 46 is shown secured to the rim of the pocket 64 via the clip 60. In the illustrated embodiment, the clip 60 is integrally formed with the body 47. In other embodiments (not shown), the clip 60 can be a separate component affixed to the body 47.

(19) The tool 46 further includes first and second planer grip surfaces 66, 68 formed at opposite top and bottom sides 70, 72 of the body 47. The grip surfaces 66, 68 are also formed in the central region of the body 47 and located adjacent the storage receptacles 58. The grip surfaces 66, 68 can be grasped by the user to hold and manipulate the tool 46.

(20) FIGS. 10A-10C illustrate a process for removing the nozzle 42 from the nozzle holder 40 using the tool 46. With reference to FIG. 10A, the removal end 48 of the tool 46 is first inserted into the tube 32 through the outlet opening 38. With reference to FIG. 10B, the annular rim 54 is moved over the nozzle 42 and pressed against the outer ring 44, causing the outer ring 44 to retract from the holding position (FIG. 10A) to the release position (FIG. 10B). With the outer ring 44 retracted, the nozzle 42 is released and ejected from the nozzle holder 40 (FIG. 10C). The ejected nozzle 42 lands within the pocket 52. The tool 46 can then be removed from the tube 32 and the nozzle 42 retrieved from within the pocket 52.

(21) FIGS. 11A-11C illustrate a process for installing the nozzle 42 into the nozzle holder 40 using the tool 46. With reference to FIG. 11A, the front portion of the nozzle 42 is inserted into the recess 56 of the installation end 50 of the tool 46, and the installation end 50 is inserted into the tube 32 through the outlet opening 38. With reference to FIG. 11B, the opposite end of the nozzle 42 is then pressed into the nozzle holder 40 until it reaches a secured position at which the outer ring 44 moves to the holding position (FIG. 11B). With the nozzle 42 secured within the nozzle holder 40, the tool 46 can then be removed from the tube 32 (FIG. 11C).

(22) Various features of the disclosure are set forth in the following claims.

Claims

1. A tool for operating a nozzle holder, the nozzle holder configured to selectively and removably couple a spray nozzle to a spray device, the tool comprising: an elongated body extending between a first end and a second end opposite the first end, the second end defining a recess configured to selectively receive and hold a portion of the spray nozzle; and a nozzle storage receptacle formed in the elongated body between the first end and the second end, the nozzle storage receptacle configured to selectively receive and hold the spray nozzle; wherein the nozzle storage receptacle defines an aperture that extends through the elongated body between a first lateral side of the elongated body and a second lateral side of the elongated body opposite the first lateral side.
2. The tool of claim 1, wherein the first end defines an annular rim and a pocket that communicates

with the annular rim, the annular rim being configured to selectively engage the nozzle holder to release the spray nozzle from the nozzle holder.

3. The tool of claim 1, further comprising a flexible arm extending from the elongated body.

4. The tool of claim 1, further comprising a pair of planar grip surfaces formed on the elongated body between the first end and the second end.

5. The tool of claim 4, wherein the planar grip surfaces are located adjacent the nozzle storage receptacle.

6. The tool of claim 1, wherein the recess includes a hexagonal shape that corresponds to a hexagonal shape of the portion of the spray nozzle.

7. The tool of claim 1, wherein the nozzle storage receptacle is further configured to receive the spray nozzle inserted from the first lateral side, and wherein the nozzle storage receptacle is further configured to receive the spray nozzle inserted from the second lateral side.

8. A tool for operating a nozzle holder, the nozzle holder configured to selectively and removably couple a spray nozzle to a spray device, the tool comprising: an elongated body extending between a first end and a second end opposite the first end, the second end defining a recess configured to selectively receive and hold a portion of the spray nozzle; and a nozzle storage receptacle formed in the elongated body between the first end and the second end, the nozzle storage receptacle configured to selectively hold and receive the spray nozzle inserted from a first lateral side of the elongated body and an opposite, second lateral side of the elongated body.

9. The tool of claim 8, wherein the nozzle storage receptacle is a first nozzle storage receptacle, and the tool further includes a second nozzle storage receptacle formed in the elongated body between the first end and the second end and configured to selectively receive and hold the spray nozzle.

10. The tool of claim 8, wherein the first end is configured to engage the nozzle holder to release spray nozzle.

11. The tool of claim 10, wherein the first end defines a pocket that does not communicate with the recess, and wherein the pocket is configured to receive the spray nozzle.

12. The tool of claim 11, wherein the first end includes an annular rim that communicates with the pocket.

13. The tool of claim 8, wherein the recess includes a hexagonal shape that corresponds to a hexagonal shape of the portion of the spray nozzle.

14. The tool of claim 8, further comprising a pair of planar grip surfaces formed on the elongated body between the first end and the second end, wherein the planar grip surfaces are located adjacent the nozzle storage receptacle.

15. A tool for operating a nozzle holder, the nozzle holder configured to selectively and removably couple a spray nozzle to a spray device, the tool comprising: an elongated body extending between a first end and a second end; an annular rim defined by the first end, the annular rim being configured to selectively engage the nozzle holder to release the spray nozzle; and a nozzle storage receptacle formed in the elongated body between the first end and the second end, the nozzle storage receptacle configured to selectively hold and receive the spray nozzle inserted from a first lateral side of the elongated body and an opposite, second lateral side of the elongated body.

16. The tool of claim 15, wherein the nozzle storage receptacle is a first nozzle storage receptacle, and the tool further includes a second nozzle storage receptacle formed in the elongated body between the first end and the second end and configured to selectively receive and hold the spray nozzle.

17. The tool of claim 15, wherein the first end defines a pocket that communicates with the annular rim, the pocket being configured to receive the spray nozzle.

18. The tool of claim 15, wherein the second end defines a recess configured to selectively receive and hold a portion of the spray nozzle.

19. The tool of claim 18, wherein the recess includes a hexagonal shape that corresponds to a hexagonal shape of the portion of the spray nozzle.

20. The tool of claim 15, further comprising a pair of planar grip surfaces formed on the elongated body between the first end and the second end, wherein the planar grip surfaces are located adjacent the nozzle storage receptacle.
